



National Highways AGS Data Visualisation Guidance

Version 0.1, July 2026

Arup-Aecom

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1. Heading

Notes: Set scope, objective, reference to pilot and value now.

2. Overview of toolkit

This guide summarises the tools and plugins that support AGS visualisation, export, and interpretation in QGIS for engineering users.

2.1 Overview

The workflow around AGS data in QGIS is not usually handled by one single plugin. Instead, different tools support different stages: - preparing AGS data for mapping - visualising boreholes and related information - exporting structured tables for reporting and dashboards - reviewing geoscience information in a GIS environment

This repo contains one part of that workflow, and external plugins can support the rest.

2.2 1. AGS CSV Export and Power BI Workflow

This repo provides the AGS processing and export tools used to move data from AGS into a structured format for reporting and dashboard use.

What These Tools Do

The tools in this repo support: - AGS to database conversion - AGS to CSV export - database to CSV export

This allows a workflow such as:

1. Start with an AGS file.
2. Convert it into a GeoPackage for use in QGIS.
3. Review or edit spatial and tabular data.
4. Export to CSV.
5. Load the CSVs into Power BI.

Why This Matters for Engineers

This workflow helps teams: - review borehole and test data spatially - preserve data lineage through export steps - standardise tables before reporting - feed dashboards and repeatable reports

Repo Tools Included

The main tools available from this repo are: - AGS to DB - AGS to CSV - DB to CSV

These appear in the QGIS Processing Toolbox when the plugin is installed correctly.

Space for Project-Specific Notes

Add your own notes here about: - how your team uses the CSV exports - which tables matter most for reporting - any naming standards for export folders - how data is handed over into reporting workflows

2.3 2. Power BI Template

This section is intentionally left open for your project-specific content.

Space for Your Input

Add details here about: - the Power BI template you have created - how users should connect the exported CSV folder - which tables are required - any expected relationships between tables - any filters, visuals, or dashboard pages already prepared

2.4 3. QGIS Plotly Graphing Plugin

This is listed as a locally developed or colleague-developed tool for graphing and visualisation inside QGIS.

Likely Use Cases

Depending on how it has been designed, a Plotly-based graphing plugin may help users: - plot test values against depth - create interactive charts from borehole or sample data - review distributions of results visually - compare locations or sample groups

Space for Colleague Input

Add colleague-provided details here about: - what the plugin is called - how it is installed - which datasets it expects - example chart types it supports - whether it reads directly from AGS, GeoPackage, or CSV - any known limitations or recommended workflows

2.5 4. OpenLog Plugin

OpenLog is an external QGIS plugin that can be used to visualise borehole logs.

Why It Is Useful

For engineering teams, it can help with: - viewing borehole log information in a more familiar log-style format - reviewing sequences down depth - presenting borehole information visually rather than only in tables

Typical Role in Workflow

OpenLog is generally useful after spatial data has already been organised in QGIS and you want a borehole-log style presentation layer.

What Users Should Check

Before relying on it, users should confirm: - it is compatible with the installed QGIS version - it supports the fields available in the current dataset - the borehole structure expected by OpenLog matches the exported or edited data

2.6 5. Geoscience Plugin

The Geoscience plugin is an external QGIS plugin used more broadly for geological and geoscience visualisation workflows.

Why It May Be Useful

It may support: - subsurface interpretation workflows - geology-focused visualisation - cross-sections or specialised geological views - integration of borehole and geology information in a GIS setting

Typical Role in Workflow

For a geotechnical consultant, this may sit alongside the AGS tools rather than replacing them.

The AGS tools in this repo are mainly about: - structured conversion - export - data handling

The Geoscience plugin may be more useful for: - interpretation - review - communication of geological context

What Users Should Check

Users should confirm: - installation method from the QGIS plugin manager - supported QGIS versions - whether the plugin expects specific geology fields or layer structures

2.7 Recommended Combined Workflow

One possible workflow for engineering teams is:

1. Use the plugin in this repo to convert AGS into a database or CSV outputs.
2. Use QGIS to review the spatial data and attribute tables.
3. Use OpenLog or another borehole log visualisation plugin when log-style review is needed.
4. Use a Plotly-based plugin where interactive charts help interpret results.
5. Export structured CSVs for Power BI dashboarding.
6. Use the Power BI template for project reporting.

2.8 Notes for Future Expansion

This guide intentionally leaves space for team-specific additions because some tools are internal or locally developed.

Good additions later would include: - screenshots of each plugin - installation steps for each plugin - example workflows by project type - a comparison table showing when to use each plugin

2.9 Related Documents

- [USER_GUIDE.md](#)
- [QGIS Basics for Geotechnical Engineers](#)
- [DATA_FLOW.md](#)

3. QGIS general

3.1 Introduction to QGIS for Engineers

This guide introduces QGIS from the perspective of a geotechnical engineer or engineering consultant who wants to view, organise, and review AGS-related spatial data without coding.



What is QGIS?

QGIS is a free GIS application used to: - load map data - view borehole and site locations - style layers visually - inspect attribute data in tables - export data for use elsewhere - run plugins and processing tools

For geotechnical work, QGIS is useful because it lets you connect location-based information to borehole, geology, sampling, and test data.

Key QGIS Concepts

Project

A QGIS project is your working file. It stores: - which layers are loaded - how they are styled - map extent and layout - links to source data

It does not usually duplicate the original data; it points to it.

Layer

A layer is a dataset shown in QGIS.

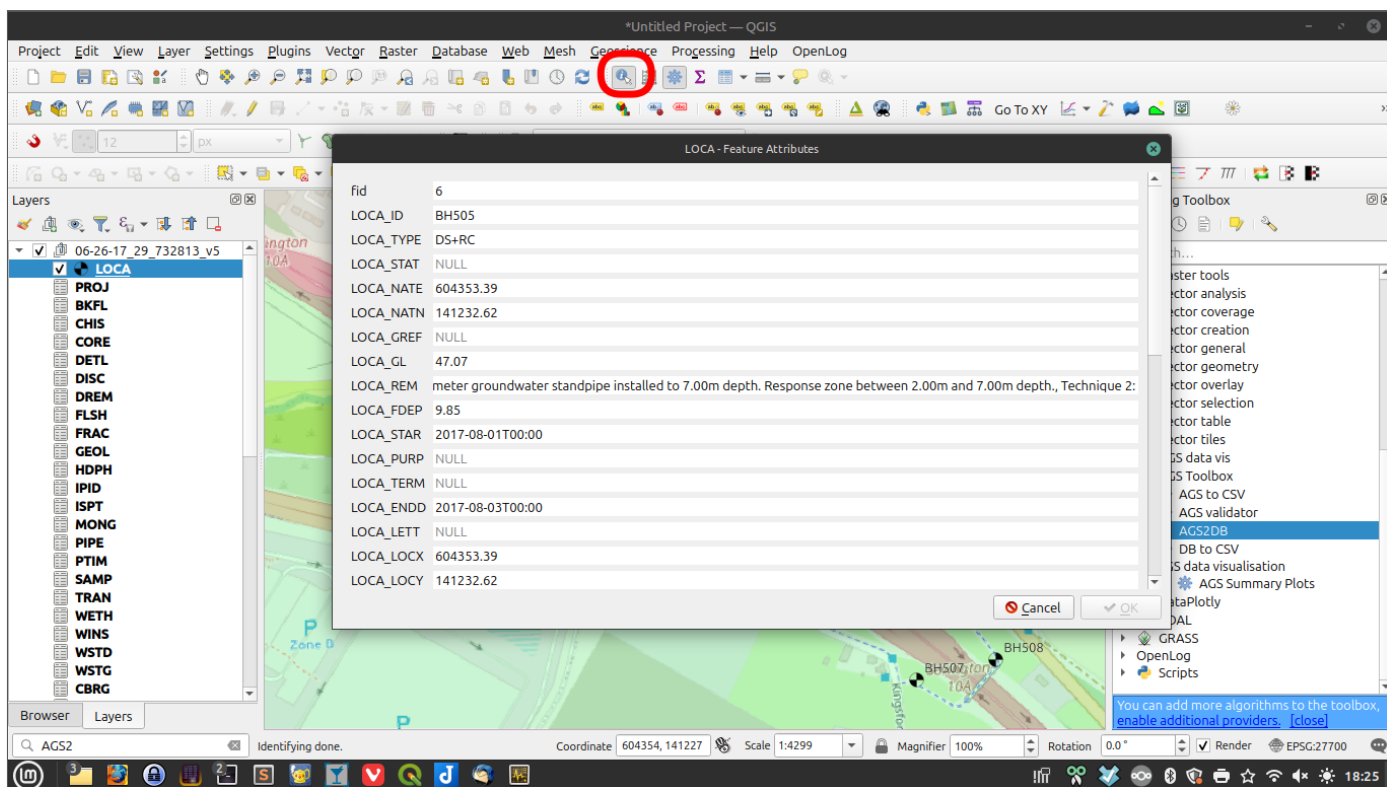
Examples: - borehole point locations - site boundary polygons - geology layers - CSV tables joined to mapped data - aerial imagery or basemaps



Tip

Think of layers as transparent sheets stacked on top of each other.

Features on active layers can be identified by selecting 'identify feature' then clicking on an item on the map.



Select 'Identify Features' (circled red) after ensuring the required layer is active.

Attribute Table

Each layer can have an attribute table. This is where the non-map data sits.

GEOL — Features Total: 279, Filtered: 279, Selected: 0

	fid	LOCA_ID	GEOL_TOP	GEOL_BASE	GEOL_DESC	GEOL_LEG	GEOL_GEO1	GEOL_GEO2	GEOL_GEO3
60	60	BH505	5.3	5.45	Firm to stiff g...	303	HY	SILT	
61	61	BH505	5.45	6.48	Probably stro...	804	HY	LIMESTONE	
62	62	BH505	6.48	6.9	Firm bluish gr...	202	AC	CLAY	
63	63	BH505	6.9	7.05	Weak bluish ...	801	AC	MUDSTONE	
64	64	BH505	7.05	9.85	Stiff bluish gr...	202	AC	CLAY	
65	65	BH506	0	0.2	MADE GROU...	102			
66	66	BH506	0.2	0.55	POSSIBLE MA...	102			
67	67	BH506	0.55	1.5	Probably very...	804	HY	LIMESTONE	
68	68	BH506	1.5	4	Probably very...	207	HY	CLAY	

Show All Features

Example attribute table (GEOL)

For boreholes, the attribute table may include: - borehole ID - coordinates - ground level - depth - project reference - geology summary

In practice, the map shows where something is and the attribute table shows what it is. Attributes can be created and edited.

Symbology

Symbology controls how layers look.

You can change: - point colour - symbol shape - label text - line style - fill colour

This helps engineers quickly distinguish between: - borehole types - sample types - geology classes - investigation status

Coordinate Reference System

The coordinate reference system, or CRS, tells QGIS how coordinates are interpreted.

For UK geotechnical work, this is often: - British National Grid: EPSG:27700

Some outputs or web maps may use: - WGS84: EPSG:4326



Tip

If coordinates appear in the wrong place, CRS is one of the first things to check.

Main Areas of the QGIS Screen

Layers Panel

Usually shown on the left. This lists all loaded layers.

You can: - turn layers on and off - reorder layers - group layers - rename layers for clarity

Map Canvas

This is the main map window. It shows the spatial view of your data.

You can: - zoom in and out - pan around site locations - click features to inspect them

Browser Panel

Used to browse files, folders, GeoPackages, and database connections.

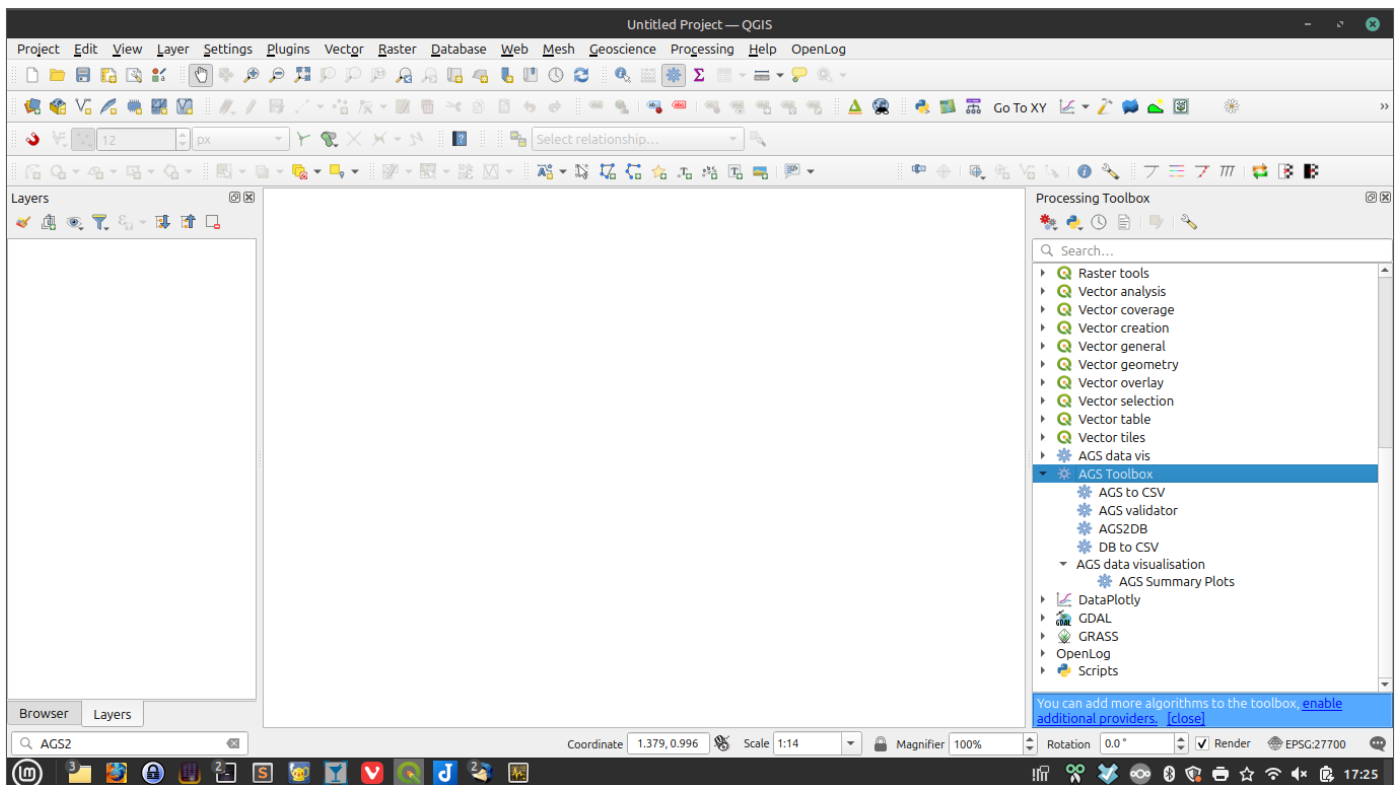
This is helpful when opening: - GeoPackages created from AGS files - folders of CSV outputs - project files

Processing Toolbox

This is where many plugins and automation tools appear.

In this repo, the AGS tools are accessed from the Processing Toolbox.

Typical use: - open Processing Toolbox - search for a tool by name - run the tool through a dialog box



Processing Toolbox is on the right hand side of the screen

Typical QGIS Tasks for a Geotechnical User

Load a spatial dataset

You might load: - a GeoPackage containing borehole locations - a site boundary shapefile - an aerial image - a terrain layer

Inspect borehole records

You can: - click a borehole on the map - open its attribute table - filter records by location or ID

Style boreholes by type or status

You might colour points by: - exploratory hole type - status - geology category - cluster or project zone

Join tabular data

QGIS can connect map layers to related tables using IDs such as: - LOCA_ID - sample IDs - project IDs

This is useful when combining mapped boreholes with exported AGS tables.

Export data

You can export selected layers or tables for use in: - Power BI - Excel - other GIS software

How Plugins fit In

Need a section on the plugin manager

QGIS plugins add extra tools to the base software.

For AGS and geotechnical workflows, plugins can help with: - AGS conversion - data export - charting - borehole log display - geoscience interpretation

Some plugins are built into your organisation's workflow. Others are external community plugins.

Good Habits for Non-Coder Users

- Keep raw AGS files unchanged and work from copies.
- Name output folders clearly by site or batch.
- Save QGIS projects regularly.
- Check layer CRS when data looks misaligned.
- Open attribute tables to confirm values before exporting.
- Keep a simple folder structure for raw data, GIS outputs, and reporting outputs.

Suggested Learning Path

1. Learn how to open a QGIS project and turn layers on and off.
2. Learn how to open an attribute table and filter rows.
3. Learn how to use the Processing Toolbox.
4. Learn which AGS-related plugins support your workflow.

3.2 Output Layouts in QGIS

Layouts are a built-in feature of QGIS they provide the means to create a report-quality output showing the map and/or other items such as tables, DataPlotly Graphs and images.

Layouts use a templating system so that a template such as a standard drawing frame can be used for many different outputs.

For Geotechnical outputs that would typically be found in reports, the types of outputs would be A4 portrait site location and parameter vs depth/elevation graphs and A3+ landscape or portrait exploratory hole location plans.

The 'QGIS National Highways package' comes with a few templates and it is expected that these will be adopted and extended by National Highways staff.

It should be noted that whilst the Layouts are accessed from within QGIS, the items such as images can be from any application or source. Images could include photographs, charts from PowerBI or Excel etc. etc. The primary reason for using QGIS layouts is to ensure consistency or report-quality output.

It is recommended that the reader familiarises themselves with Layouts by reading the QGIS documentation [here](#) or watch a [video here](#)

3.3 QGIS useful plugins

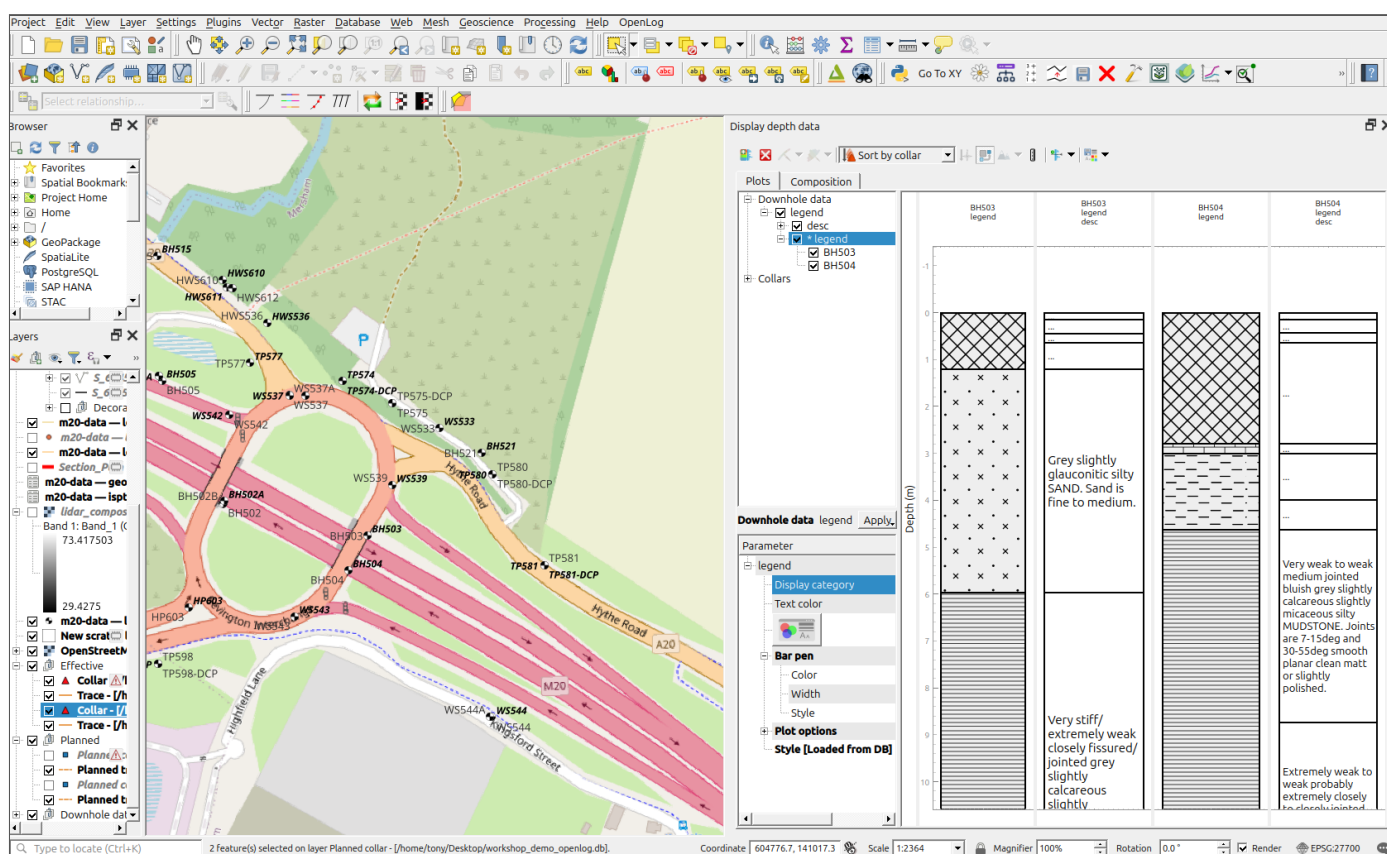
Openlog



TD This process needs to be tested by someone new to it.

OpenLog is a freely available QGIS plugin that enables the user to view legends, soil descriptions and graphical data with depth. It is extremely useful for quickly viewing exploratory hole information. It is **not** intended to be a way of presenting logs in a report.

A sample output is shown below. The soil descriptions are dynamic, in that they become visible as the user zooms into the log.



Screenshot of QGIS showing the OpenLog graphical logs and soil descriptions.

There is very useful guidance available via the [OpenLog user guide](#)

Note

The OpenLog guidance is not reproduced here because the official guidance will always be more up to date. However please ignore the video at the top of the official guidance page. It refers to a database type that is not recommended.

OpenLog requires mapping of the GeoPackage data to their specific spatialite database. This is not a complicated process, but it does involve several steps.

1. Create a spatialite database [help](#)

 Note

Use the [spatialite option](#) , not the xplordb option when creating a database

1. After creating the spatialite database, [connect to it](#) and
2. add data by following the guidance [here](#)
3. Import tabular [downhole data](#)

 Note

There is no need to add survey data.

 Tip

Select 'depth' for Domain if in doubt.

Select 'extended' for geology information and 'discrete' for test information e.g. SPT.

Add as much information as you wish, but note that space on a screen is limited, so keep it as simple as possible.

 Tip

In the Column definition part of the Downhole data import screen there are data type options. Normally these will be one of three: 1. 'NOMINAL' - e.g. soil descriptions (GEOL_DESC)

1. 'CATEGORICAL' e.g. Legend codes (GEOL_LEG) or
2. 'NUMERICAL' e.g. SPT test results (SPT_NAL)

Once the data has been imported the logs are [ready for viewing](#)

The legend graphics can be changed using the functionality shown [here](#)

 Note

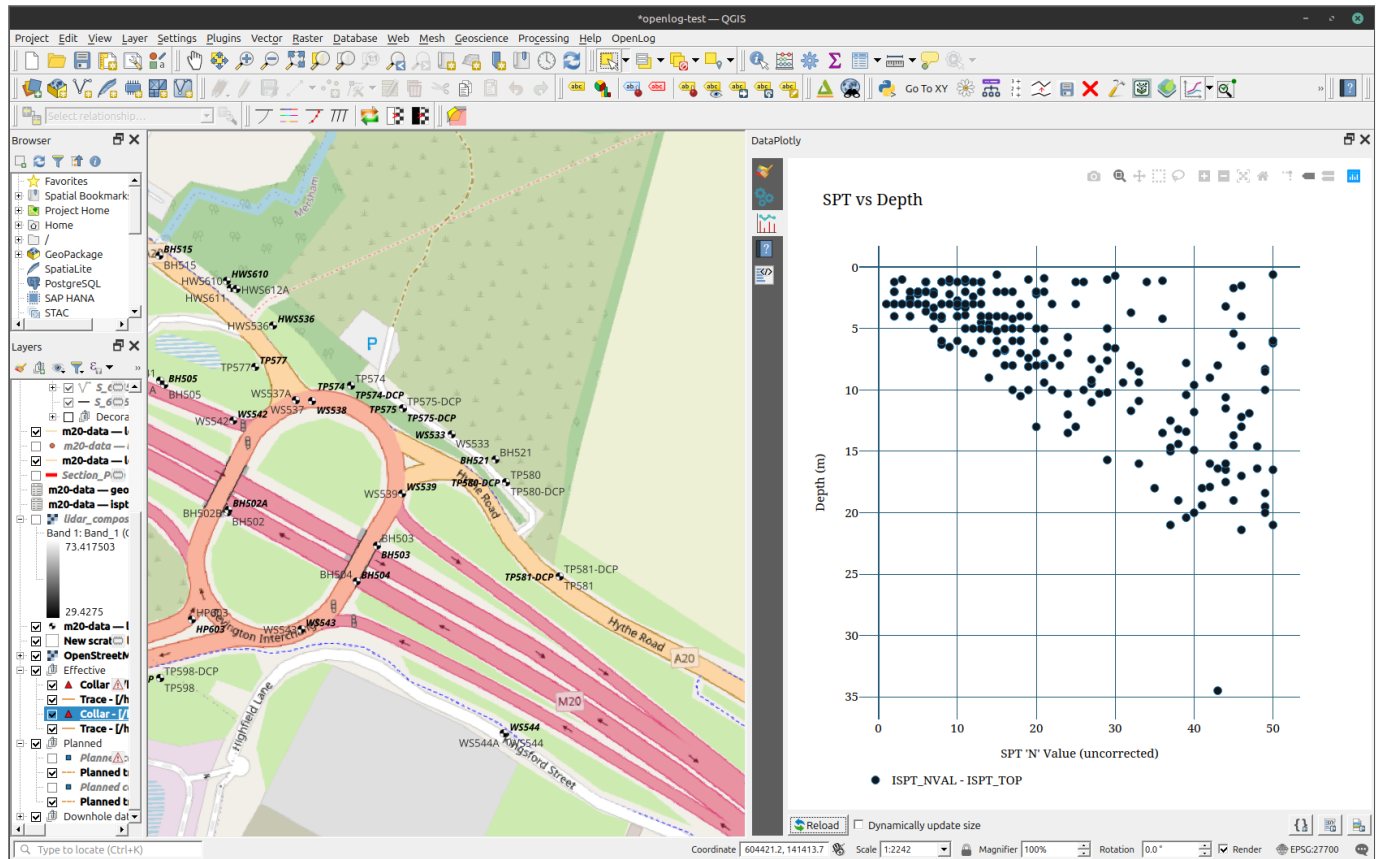
The developers of OpenLog have made an [AGS file-friendly interface](#), however this is part of their Premium (paid-for) offering. The use of the GeoPackage file created using the ags-tools plugin offers a very similar experience.

DataPlotly

TD This process needs to be tested by someone new to it.

DataPlotly is a freely available QGIS plugin that enables the user to view simple charts such as SPT vs Depth. It is extremely useful for quickly viewing filtered, exploratory hole test data. It can be used, with the QGIS Layout functionality to produce report-quality output.

An example plot of SPT vs Depth is shown below



DataPlotly is very simple to use and has good guidance [here](#)

It can create the following, common plot types:

- Scatter
- Histogram
- Bar
- Box
- Ternary
- Pie



Tip

In order to be able to use the 'Use only selected features' functionality when tables other than LOCA are being used from the GeoPackage data source, 'relations' must be created between LOCA and the data table and the 'SelectbyRelationship' plugin must be selected. Guidance on how to do this is explained [here](#). [ins link](#)

 Warning

Avoid using multiple plots if the Y axis is inverted (i.e. plot vs depth). There is a [known bug](#) whereby the layout is not honoured

Relations

The Geopackage created with the ags-tools plugin comprises a spatial table of the LOCA group data and other non-spatial tables that hold the data of all the other groups, e.g. GEOL and ISPT.

The main element of QGIS is the map interface and the ability to select and filter data based on its location. This is fine for the spatial table LOCA because it has co-ordinates and the markers of the the exploratory holes can be presented on the map.

However for AGS groups that don't have co-ordinates e.g. GEOL and ISPT it is not possible to select them via the map. A workaround would be to assign co-ordinates to each row of the table when the GeoPackage is created, however this would create a very cluttered layer tree and map.

The better way to handle non-spatial table is to use the QGIS functionality known as '**Relations**'.

This provides the ability to create relationships between spatial and non-spatial tables using a common data item, which is typically the LOCA_ID.

Once the relations have been created selecting an exploratory hole will also select the data linked by the relationship. This is particularly useful when using the DataPlotly plugin and when filtering or highlighting data using Attribute tables.



Note

In order to use this related data the 'Select by Relationship' plugin must be installed.

The following guidance provides a step by step process of creating and using 'relations' to create a plot of SPT vs depth for a selection of exploratory holes.

4. Data preparation

In order to make the most of the ags-tools suite of applications the AGS data file must be valid and of sufficient data quality for the type of analysis and visualisation that needs to be undertaken.

Checking the AGS file format for validity is explained [here](#).

For example if markers of all exploratory holes are to be plotted on a map, or aerial photo then 100% of the exploratory hole co-ordinated (LOCA_NATE and LOCA_NATN) must be populated, with values that reflect the real-world position of the exploratory holes and 100% of the exploratory holes must have the correct ID (LOCA_ID). However, for the purposes of this visualisation, the data started, or the sample information is not needed.

On the other hand if the user wished to create a plot of SPT vs Depth. then the exploratory co-ordinate information is not useful, but the insitu test depth and results are. (ISPT_DPTH and ISPT_NVAL)

Data quality therefore should be considered in the context of the use of the data.

Warning

GDMS only requires **75%** population of the co-ordinate data, so do not assume that the AGS data held has the level of data quality required for all purposes.

4.1 Data quality

When assessing the quality of data it is useful to consider different aspects of data that together provide a measure of data quality. These aspects are termed 'dimensions' and are:

- Accuracy
- Completeness
- Uniqueness
- Consistency
- Timeliness
- Validity
- Integrity
- Currency
- Reasonableness

The definitions are set out in the DAMA DMBOK publication.

The main dimensions that are relevant to AGS data used in analysis are: Completeness, Validity Integrity, Uniqueness and Reasonableness. Examples are given below:

Term	Example
Validity	Is the final depth of the hole (LOCA_FDEP a positive number)
Completeness	Have all the exploratory holes been assigned co-ordinates (LOCA_NATE and LOCA_NATN)
Integrity	Does the base depth of a strat match the top depth of the one below it (GEOL_BASE, GEOL_TOP)
Uniqueness	Do all the top depth of strata in a single exploratory hole have a unique value (GEOL_TOP)
Reasonableness	Does a sample described as stiff clay have a strength test that matches the description (SAMP_DESC)

 Note

The AGS file format checks for some of the dimensions, such as Uniqueness and Validity, however it only checks against the format requirements (keysets and data types) It does not check the data itself.

The AGS has created a tool to enable the dimensions to be quantified against criteria and assessed. Further information about the tool can be found [here](#)

The visualisation summary tool in the ags-tools plugin provides an immediate, graphical way of looking at the data. It can flag any obvious outliers of data (Reasonableness and/or Validity) and missing data (Completeness).

 Note

Note that the Accuracy of data can only be assessed where the real world values are known. e.g. Location ID.

5. Data visualisation workflow

5.1 Overview

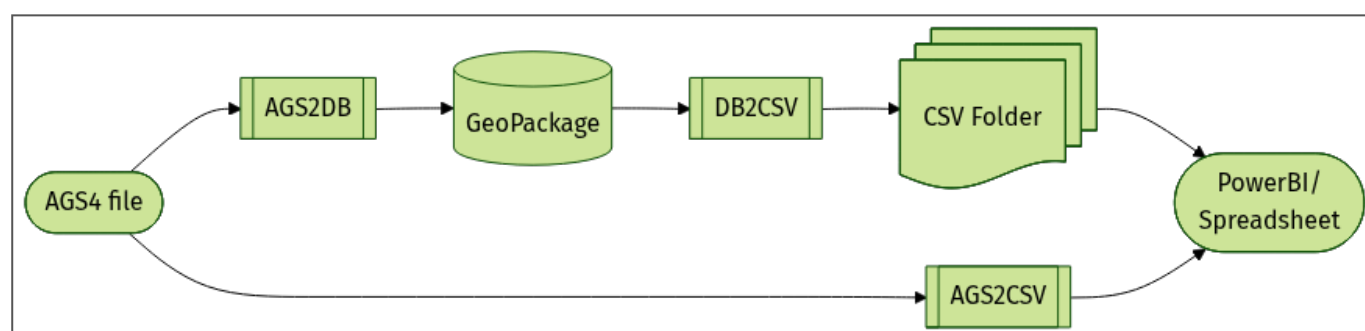
The AGS Tools plugin provides three processing algorithms for exporting geotechnical AGS4 data:

Algorithm	Source	Destination	Purpose
AGS to DB (ags_2_db_algorithm.py)	.ags file	GeoPackage / Spatialite	Parse and store AGS data in a spatial database
AGS to CSV (ags_2_csv.py)	.ags file	Folder of .csv files	Export AGS data directly to CSVs for Power BI or other data visualisation tools
DB to CSV (db_2_csv_algorithm.py)	GeoPackage / Spatialite	Folder of .csv files	Export an edited database to CSVs for Power BI

All three algorithms write a source_file column into every output row, providing a complete audit trail of where data came from. Deduplication uses a prefix-match key on that column to prevent duplicate rows when re-exporting or appending.

5.2 The Data Pipeline

Full End-to-End Flow



A typical workflow uses **AGS to DB first**, then **DB to CSV** — this allows data to be reviewed and edited in QGIS before being exported to Power BI. The **AGS to CSV** pathway skips the database entirely for quick exports.

5.3 The source_file Column

Every row in every output CSV contains a `source_file` column. Its format varies by pathway and records full data lineage:

Pathway	Format	Example
AGS to CSV	ags_filename csv_folder_name project_id	Esholt.ags site_export BH-001
AGS to DB (stored in DB)	ags_filename db_filename project_id	Esholt.ags Esholt.gpkg BH-001
DB to CSV (appended)	ags_filename db_filename project_id csv_folder_name	Esholt.ags Esholt.gpkg BH-001 site_export

The `project_id` is read directly from the `PROJ_ID` field in the AGS file's `PROJ` group. If absent, it is omitted from the string.

6. Data visualisation using AGS Tools QGIS Plugin

6.1 General overview

Introduction

This guide provides a practical, step by step workflow for geotechnical engineers and consultants to convert AGS data, build QGIS databases, prepare CSVs for Power BI, and generate geological cross sections.

You will use AGS Tools to:

1. Convert AGS files into a QGIS database for technical checking.
2. Export CSV data for Power BI.
3. Generate geological section outputs for interpretation and communication.

The focus is on what to click, what to check, and what "good" looks like from a geotechnical perspective.

Before You Start - 5-Minute Checklist

1. Install QGIS (recommended: QGIS 3.22 or later).
2. Install and enable AGS Tools.
3. Create a simple folder structure for each project:
4. 01_Raw_AGS
5. 02_QGIS_Database
6. 03_PowerBI_CSV
7. 04_Sections
8. Keep raw AGS files read-only.
9. Confirm project CRS (typically EPSG:27700 in the UK).

Screenshot Placeholder

[Insert screenshot: QGIS download page]

[Insert screenshot: QGIS home screen on first launch]

Installation/unpacking etc.

Before using any AGS conversion tool, it is worth spending a few minutes making sure the plugin is installed correctly. This prevents frustrating errors later when you are under time pressure.

Install and enable AGS Tools

1. Open QGIS.
2. From the top menu, click Plugins > Manage and Install Plugins.
3. In the search box, type AGS Tools.
4. Select AGS Tools, then click Install Plugin (or Enable if already installed).
5. Restart QGIS if prompted.

Confirm the tools are visible in Processing Toolbox

1. Open Processing Toolbox from the right-hand panel.
2. Search for AGS.
3. Confirm these tools appear:
4. AGS2DB
5. AGS2CSV
6. DB2CSV
7. DB to Cross-Sections
8. Summary plots

If these are visible, your installation is ready.

If one or more tools are missing, it is usually a plugin enablement issue rather than a data issue. Re-check installation first before troubleshooting the AGS file.

If your team provides a ZIP package ?? team??

1. Save the ZIP in a known folder.
2. Unzip fully into a normal folder (do not open and run from inside the ZIP).
3. Keep version names clear, for example AGS_Tools_v1_2.
4. If required by your team process, copy the unpacked plugin folder into your QGIS plugins directory.

Good practice

- Use one approved plugin version per project issue.
- Avoid combining outputs from different plugin versions in the same final issue pack.

Once this check is complete, you are ready to create your first review database from AGS.

Screenshot Placeholder

[Insert screenshot: Plugins > Manage and Install Plugins window]

[Insert screenshot: Processing Toolbox showing AGS tools]

Appendix - Plain-language tool summary

- AGS2DB: Converts AGS into a QGIS database for engineering review.
- AGS2CSV: Quick direct export from AGS to CSV.
- DB2CSV: CSV export from reviewed database content.
- DB to Cross-Sections: Generates geological sections from the database.

6.2 Validate AGS files

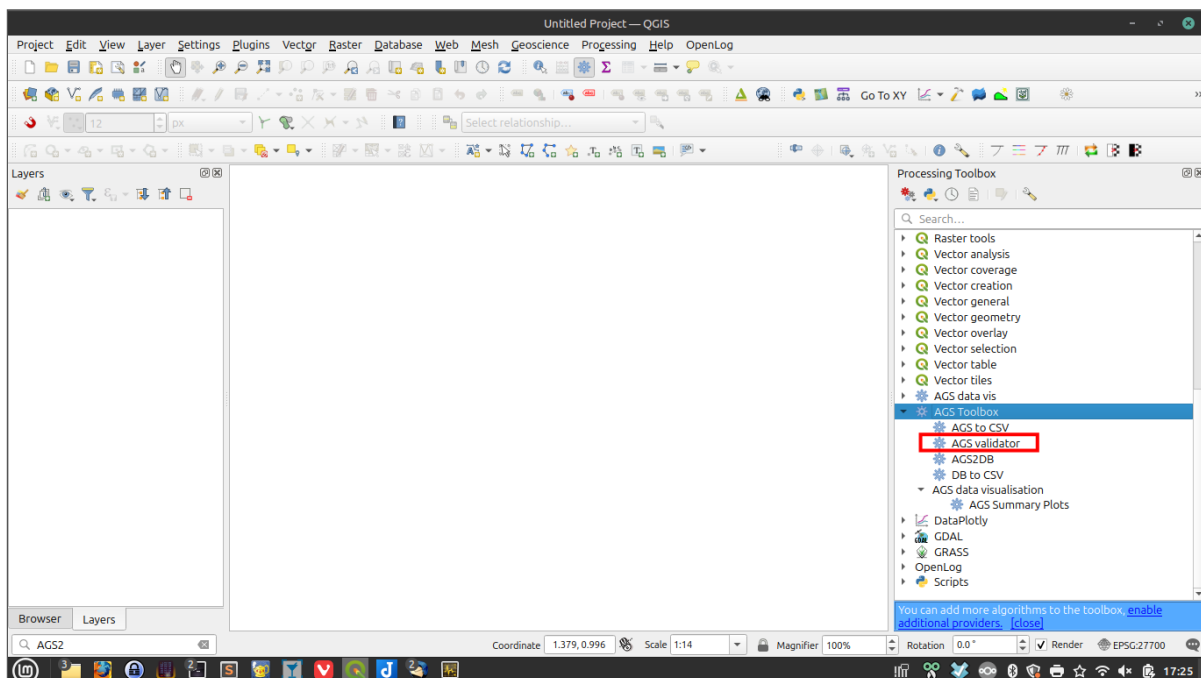
An AGS file can be validated from within QGIS, using the ags-tools plugin.

Tip

If the ags-tools validator fails for any reason, other validators are available including the one hosted by the BGS [here](#).

Tip

If you can't find the processing toolbox panel because it has been closed, search for 'AGS validator' using the search box in the bottom left hand corner of the QGIS application.



Launch AGS4 validator

ags-tools: AGS4 validator dialog

The following items are essential:

- **'Input File'** = AGS filename [input]
- **'Coordinate Reference System'** This will be filled in by default to the OSGB grid. This must not be changed.
- **'Checkers'** Normally the default 'ags' should be selected.

The following items are optional:

- **'Output File'**

6.3 Create database from AGS (AGS2DB)



AGS4 files must be validated before conversion. See [ags tools validator](#) [Insert a link]

AGS4 Files must be converted to GeoPackage files to be viewed on a map in QGIS. This can be done by using the ags-tools processing plugin.

Use AGS2DB to create a spatial, reviewable database before reporting.

This is usually the best first technical step in a geotechnical workflow because it gives you a map view and table view of the same dataset. In practice, this is where you catch location issues, ID mismatches, and obvious depth anomalies early.

In QGIS select the ags-tools processing plugin from the processing toolbox panel and select AGS2DB



If you can't find the processing toolbox panel because it has been closed, search for 'AGS2DB' using the search box in the bottom left hand corner of the QGIS application.

The following form is displayed:

AGS2DB data entry dialog

The following items are essential:

- **'Input File'** = AGS (*.ags) filename
- **'Coordinate Reference System'** This will be filled in by default to the OSGB grid. This must not be changed.

The following items are optional:

- **'Output File'** If this isn't specified a file will be written to a 'temporary' location.
- **'Initials or name of of the user'**. This is used to record the metadata in the source table of the GeoPackage.
- **'Style file'**. This is used to set the symbology of the exploratory hole markers. If none is set a default file will be used. See [QGIS: docs - symbology](#)
- **'Use same Folder as AGS file'** If no Output file is specified the output will be written to the same folder as the input file.
- **'Include Google map basemap'**. If this is ticked a Google-sourced basemap will be added to the layers.
- **'Geology and DTM maps'**. Additional layers provided mostly by the BGS (e.g. BGS Bedrock) can be added to the map. Select from the list. The order of the layers is automated, as is the transparency. These can be changed after creation.

Selecting 'Run' will process the AGS file, create and load a GeoPackage file into QGIS add chosen layers and apply a symbology to the exploratory hole markers.

What you should see after run

- A new GeoPackage file in your output folder.
- AGS tables loaded as layers/tables.

- Borehole point layer (for example LOCA) where coordinates exist.
- source_file field present for data lineage.

GeoPackage description

A GeoPackage is a database file. The format is set by the OGC([include link](#)) The AGS conversion process creates a table LOCA that includes the geometry of exploratory holes. All the other groups in the AGS file are converted into tables in the GeoPackage. Additionally the AGS UNIT and TYPE heading rows are included as a single table 'UNTY'. Two other tables are also created: 'source' and 'version_control'. These tables are included to enable a user to understand the source of the data and manage any changes to the GeoPackage, such as the addition of geology codes (GEOL_GEOL).

Geotechnical QA checks to perform immediately

1. LOCA map check: exploratory holes appear in correct site area.
2. ID check: LOCA_ID labels are accurate.

Common mistakes to avoid

- Running with the wrong AGS file revision.
- Saving output into the raw input folder.
- Skipping map checks

When the database checks are complete the data can be used for analysis, either within QGIS or exported to CSV for use in PowerBI or spreadsheets.

- Use AGS2CSV for fast direct export.
- Use DB2CSV for controlled export after review.
- In QGIS tools such as Attribute tables, DataPlotly and OpenLog can be used to review change and analyse the data.

Screenshot Placeholder

[Insert screenshot: AGS to Database tool dialog with fields highlighted]

[Insert screenshot: LOCA layer displayed on map]

[Insert screenshot: GEOL table in attribute table view]

6.4 Create CSV for PowerBI from AGS (AGS2CSV)

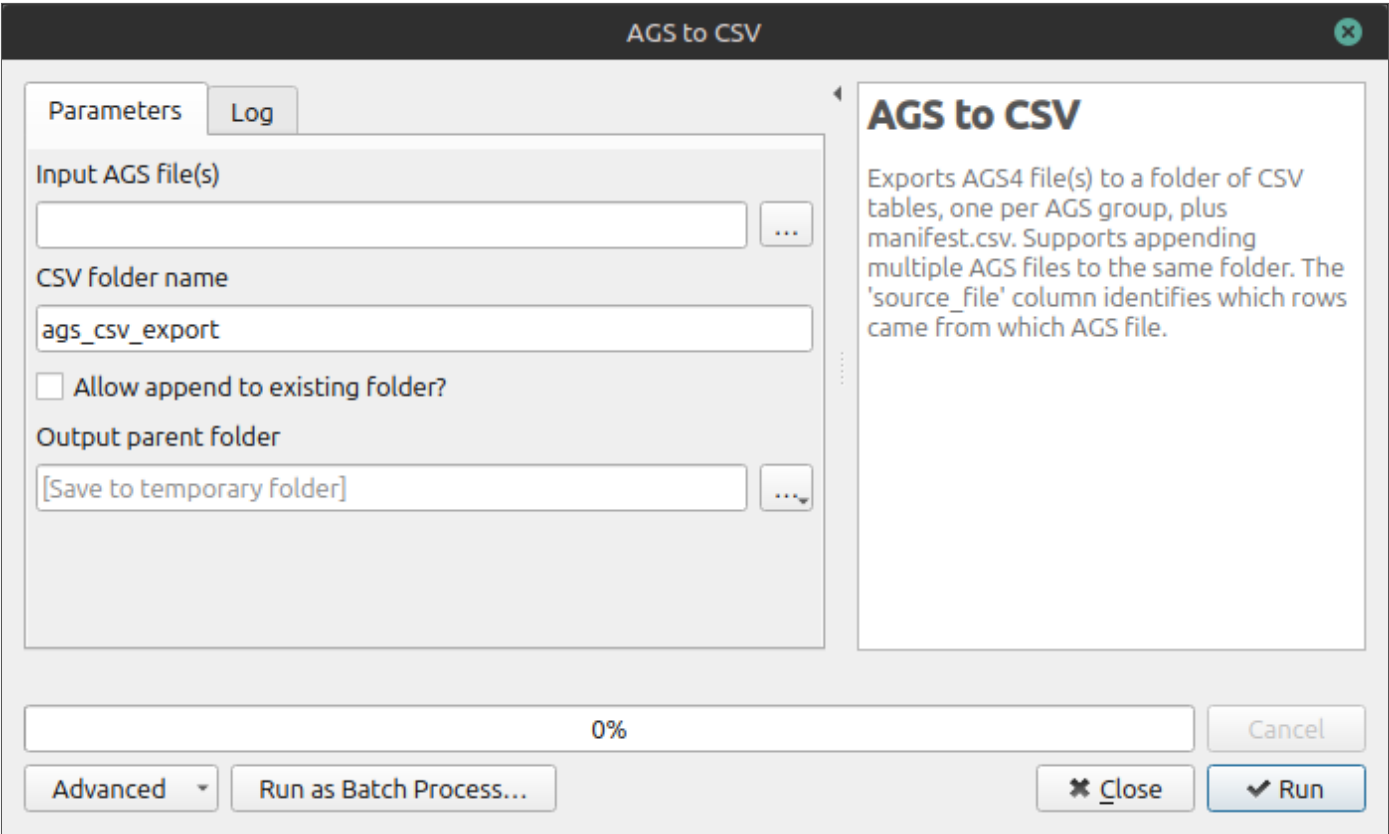
Use this when you need a direct AGS to CSV export quickly.

This route is useful when you need a quick data drop for early visualisation. It is fast, but because it bypasses database review, it should be treated as a draft-first route rather than a final issue route.

Best use cases

- Early-stage data exploration.
- Rapid draft checks.
- Projects where no QGIS edits are needed.

Step-by-step UI guidance



AGS2CSV dialog

The following items are essential:

- **'Input AGS File(s)'** = AGS (.ags) filename.
- **'Output parent folder'** All the CSV files will be created in a single folder.

The following items are optional:

- **'Allow append to existing folder'** Use if multiple AGS projects are to be exported for analysis at the same time.

Click Run.

If you are producing multiple drafts in one day, use clear folder names that include issue stage and date. This makes handover and rollback much easier.

Expected output

- One CSV per AGS group that exists in the source.
- manifest.csv (what exported and row counts).
- ags_column_metadata.csv (column/unit metadata).

Important

- Only existing source groups and columns are exported.
- Placeholder tables are not created for missing groups.

Screenshot Placeholder

[Insert screenshot: AGS to CSV dialog]

[Insert screenshot: Export folder showing CSV outputs and manifest.csv]

6.5 Create CSV for PowerBI from database (DB2CSV)

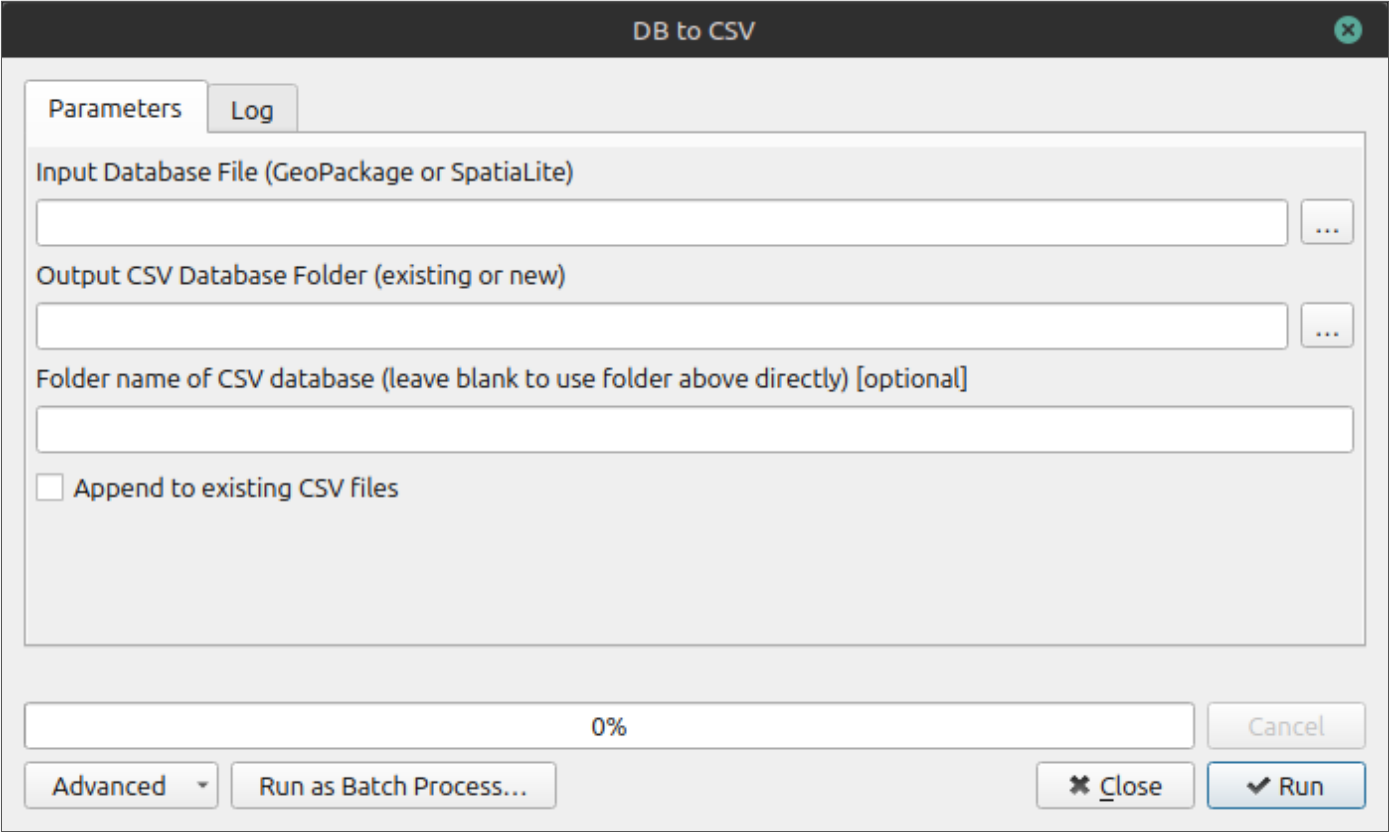
DB2CSV provides the ability to create a suite of CSV files from a GeoPackage database. This is useful if the data in the AGS file has been changed in QGIS, for example if Geology Codes have been added and the user wants to analyse the data in an application that uses CSV for data input, such as spreadsheets or a business analytics platform such as PowerBI or Tableau. This is also useful if multiple AGS file data sets need to be analysed together.

This is the recommended export route. It gives the user confidence that the CSV output reflects reviewed data and not just raw incoming AGS content.

Why this route is often preferred

- You are exporting from reviewed data.
- It supports better QA and audit traceability.
- It is usually the safer route for formal issue.

Open Processing Toolbox. click on DB2CSV.



ags-tools: DB2CSV dialog

The following items are essential:

- **'Input Database File'** = GeoPackage (.gpkg) filename.
- **'Output CSV Database Folder'** All the CSV files will be created in a single folder.

The following items are optional:

- **'Folder name of CSV database'** This is a file name of the zipped CSVs
- **'Append to existing CSV files'** Use if multiple AGS projects are to be exported for analysis at the same time.

Click 'Run'

Expected output

- One CSV per AGS group that exists in the source.
- manifest.csv (what exported and row counts).
- ags_column_metadata.csv (column/unit metadata).

Important

- Only existing source groups and columns are exported.
- Placeholder tables are not created for missing groups.

[Insert screenshot: Database to CSV dialog]

[Insert screenshot: CSV output folder for final issue]

6.6 Visualisation summary charts in QGIS

Include summary chart workflow

6.7 Geological sections in QGIS

Use this workflow to generate cross-sections from your reviewed database.

This part of the workflow is where geotechnical interpretation becomes much more visual. A good section setup helps the team communicate strata relationships, depth trends, and ground model assumptions clearly.

Purpose

DB to Cross-Sections creates geological section outputs for interpretation, design discussions, and reporting.

Typical geotechnical workflow

1. Complete AGS2DB.
2. Review LOCA and GEOL content.
3. Run DB to Cross-Sections.
4. Define section line.
5. Review output visually.
6. Export PNG for issue, if required.

Step-by-step UI guidance (DB to Cross-Sections)

1. Open Processing Toolbox.
2. Run DB to Cross-Sections.
3. Set Input DB to your GeoPackage.
4. Enter a clear section name, for example A-Ap.
5. Define Section Line using one method:
6. Leave blank for automatic line generation.
7. Enter coordinates as sx,sy,ex,ey.
8. Use interactive draw on map.
9. Set Section width buffer (m).
10. Optional: set LiDAR raster for ground profile.
11. Optional: set PNG report export.
12. Click Run.

If you are new to section work, run a first pass quickly, then refine the section line and buffer. Iteration is normal and usually improves the final output significantly.

Practical interpretation tips

- Start with a broader buffer (500 m to 1000 m) and refine.
- Keep section names aligned with your drawing references.
- If the section is empty, first verify LOCA locations and GEOL depth data.
- If orientation is not useful, reverse or redraw the line.

It is good practice to save at least one preview PNG during drafting so project reviewers can comment on section positioning before final issue.

Common issues and fixes

1. No section features:
2. Check LOCA and GEOL records exist.
3. Increase buffer width.
4. Ensure the section line crosses borehole extents.
5. Ground profile missing:
6. Confirm raster file path.
7. Test without raster to isolate the issue.
8. Section looks over-exaggerated:
9. Reduce vertical exaggeration and rerun.

Screenshot Placeholders

[Insert screenshot: DB to Cross-Sections dialog]

[Insert screenshot: Section line drawing on map canvas]

[Insert screenshot: Generated section output window]

[Insert screenshot: Exported PNG preview]

7. Power BI

7.1 Introduction to PowerBI Dashboard

This guide explains how to use the **AGS Data Visualisation Dashboard** Power BI template created as part of the AGS Toolkit.

The dashboard is designed to help geotechnical engineers quickly view, check and explore AGS data without needing to manually build plots, write queries, or understand the technical structure of Power BI.

Important

The dashboard is intended to help users visualise and review AGS data more efficiently.

It does not replace:

- Engineering judgement.
- Formal geotechnical interpretation.
- Data validation by a competent engineer.
- Review of the original AGS file where required.

Users should treat the dashboard as an interactive visual aid for exploring AGS data, identifying trends and checking imported records.

Recommended Workflow

A typical review workflow is:

1. Open the Power BI template.
2. Enter the CSV folder path.
3. Check the **Project Summary** page.
4. Confirm the correct project and source file are loaded.
5. Check the map and location details.
6. Review which AGS tables contain data.
7. Use the analysis pages to inspect specific test data.
8. Apply filters where needed.
9. Check dynamic units before interpreting charts.
10. Use the tables to verify plotted values where required.

What the Power BI Dashboard Does

The Power BI dashboard automatically reads the CSV files exported from the AGS Toolkit and turns them into a set of interactive geotechnical plots, maps, tables and summaries.

Each CSV file represents one AGS table, such as `LOCA`, `GEOL`, `SAMP`, `ISPT`, `LLPL`, `TRIT`, `RUCS`, `RPLT`, `RDEN`, `GRAG` or `GRAT`.

Once the CSV folder is loaded, the dashboard automatically:

- Imports the available AGS tables.
- Applies the required data transformations.
- Links related tables together.
- Creates maps, plots and summary tables.
- Reads available units and axis labels from the AGS data where possible.
- Allows the user to filter by project, source file, location type, geological unit and borehole ID.

This means users do not need to manually format AGS tables or create Power BI charts themselves.

Before Opening the Power BI Dashboard

Before using the Power BI dashboard, the AGS file must first be converted into CSV format using the **AGS to CSV** tool in the QGIS AGS Toolkit.

The converter will create a folder containing multiple CSV files. Each CSV file stores the data from a different AGS group.

For example, the output folder may contain files such as:

- LOCA.csv
- GEOL.csv
- SAMP.csv
- ISPT.csv
- LLPL.csv
- TRIT.csv
- GRAG.csv
- GRAT.csv
- RUCS.csv
- RPLT.csv
- RDEN.csv
- ags_column_metadata.csv
- manifest.csv

Not every AGS file will contain every table. This is normal. For example, a project with no rock testing will not contain useful rock testing records, and the rock testing page may appear empty.

Important

Keep all generated CSV files together in the same folder. Do not rename, move, edit or delete individual CSV files before loading the folder into Power BI.

Screenshot Placeholder – Generated CSV Folder

[Generated CSV folder](#)

Dashboard Page format

Most pages in the dashboard follow the same basic structure.

There is usually:

- A main chart or map.
- One or more supporting tables.
- Summary cards showing key values.
- Dynamic units and axis labels.
- Filters at the bottom of the page.

The dashboard is interactive. When you apply filters, the charts and tables update automatically.

7.2 Opening the PowerBI Template

Opening the Power BI Template

Open the supplied **AGS Data Visualisation Dashboard Power BI template** file.

The template file will usually have the extension:

*.pbit

When the template opens, Power BI will ask for a parameter called **FolderPath**.

This is where you tell Power BI where the converted AGS CSV files are stored.

FolderPath

In the **FolderPath** box, enter the full path to the folder containing the CSV files.

 **Example**

C:\Projects\Example Site\AGS CSV Output\

 **Note**

You should enter the folder path only. Do not select an individual CSV file.

After entering the folder path:

1. Click **Load**.
2. Power BI will read the CSV files in the folder.
3. The dashboard will refresh.
4. The report pages will populate with the imported AGS data.

Depending on the size of the AGS dataset, the first load may take a few moments.

Screenshot Placeholder – FolderPath Input Window

Paste screenshot here:

Power BI FolderPath input

7.3 PowerBI General Features

Using Filters

The filters at the bottom of each page allow you to focus the dashboard on specific parts of the AGS dataset.

Common filters include:

- **Project ID** – filters the report to a specific project.
- **Location Cluster** – filters by location cluster if this field is present in the AGS data.
- **Source File** – filters by the original AGS source file.
- **Location Type** – filters by investigation type, such as CP, DP, RC, WS or other recorded hole types.
- **GEOL_GEOL** – filters by the primary geology code.
- **GEOL_GEO2** – filters by the interpreted geology or second geology code.
- **Location ID** – filters by individual borehole or investigation location.

Filters can be used to narrow the dashboard down to one borehole, one geological unit, one investigation type or one source file.

For example, you may want to:

- View only one borehole.
- Compare SPT results in clay only.
- Check all records from one source AGS file.
- View only rotary core holes.
- Look at PSD curves from a small number of selected locations.



Tip

If a chart is very busy, apply filters to reduce the number of records shown. This is especially useful on pages such as Particle Size Distribution, where many curves can overlap.

Dynamic Units and Axis Labels

A key feature of the dashboard is that units and axis labels are shown dynamically where possible.

This means the dashboard reads unit information from the imported AGS data, rather than relying only on fixed hard-coded labels.

This helps users understand:

- What measurement is being plotted.
- Which unit is being used.
- Whether the displayed values match the source AGS file.
- Which axis has been selected on charts where the user can switch between depth and elevation.

Units are shown in small units tables or panels on the report pages.

For example, a plot may show:

- X-axis: N Value
- Y-axis: Elevation (mAOD)

or:

- X-axis: Liquid Limit (%)
- Y-axis: Plasticity Index

This is useful because AGS files can vary between projects, and users need to be confident that values are being interpreted with the correct units.

7.4 Available Geotechnical Pages

The following sections explain each page in the template and how to use it. Use the index on the right to jump to the area of interest.

Project Summary Page

The **Project Summary** page is the first page shown after the AGS CSV folder has been loaded.

This page gives an overview of the imported dataset and helps confirm that the correct project has been loaded.

It answers four important questions:

1. Which project has been loaded?
2. Which AGS source file has been imported?
3. Where are the investigation locations?
4. Which AGS tables contain data?

Screenshot Placeholder – Project Summary Page

Paste screenshot here:

[Project Summary page](#)

Folder Name / Report Title

The large title at the top of the page is generated from the selected folder name.

This helps users quickly confirm which AGS CSV folder is currently being visualised.

Project Information Table

The project information table summarises key project metadata from the AGS data.

It typically includes:

- **Project ID**
- **Project Name**
- **Location**
- **Source File**

This provides a quick check that the correct AGS file and project have been loaded.

Borehole Location Map

The map shows the investigation locations contained in the AGS dataset.

Each point represents a borehole or investigation location from the `LOCA` table.

The map uses latitude and longitude values generated from the northing and easting coordinates during the AGS to CSV conversion process.

The points are coloured by **Hole Type** or **Location Type**, allowing users to distinguish between different investigation methods, such as cable percussion, dynamic probing, rotary core, window sample or other recorded types.

Users can:

- Zoom in and out.
- Pan around the site.
- Hover over points to view location information.
- Select points to filter related information on the page.

AGS Data Loaded Table

The **AGS Data Loaded** table shows which AGS groups were imported from the CSV folder.

For each AGS group, it displays information such as:

- AGS group name.
- Number of rows imported.
- Whether the table contains data.
- Export date.
- Source file.

This table is useful for checking whether the expected AGS tables are present.

For example, if the SPT page is empty, the user can check whether the **ISPT** table contains data.

Location Details Table

The **Location Details** table provides a summary of the investigation locations.

It typically includes:

- **Hole ID**
- **Hole Type**
- **Final Hole Depth**
- **Source File**

This allows users to quickly review which boreholes or investigation points are available in the dataset.

Report-Wide Filters

The filters at the bottom of the Project Summary page are also used throughout the report.

Selections made here can be used to focus the dashboard on a specific project, source file, hole type, geological unit or location ID.

SPT Analysis Page

The **SPT Analysis** page displays Standard Penetration Test results from the AGS data.

It allows users to review how SPT N-values vary with depth or elevation, and how those values relate to the recorded geology.

Screenshot Placeholder – SPT Analysis Page

Paste screenshot here:

[SPT Analysis page](#)

SPT N-Value Plot

The main scatter plot shows SPT results.

Each point represents one SPT record.

The points are coloured by geology, allowing users to compare SPT values across different geological units.

X-Axis Selection

The X-axis can be changed depending on what value the user wants to plot.

Available options may include:

- **N Value** – the recorded SPT N-value.
- **N60 Value** – the energy-corrected N-value, where available.

If N60 values are not present in the AGS dataset, the N60 option may not provide useful results.

Y-Axis Selection

The Y-axis can be changed to show either:

- **SPT Depth** – depth below ground level.
- **SPT Elevation** – elevation in metres above ordnance datum, where available.

Elevation is useful when comparing results across boreholes with different ground levels. Depth is useful when reviewing data relative to the ground surface.

Summary Cards

The summary cards provide quick statistics for the currently selected data.

They typically show:

- **Average N** – the average SPT N-value.
- **Number of SPT Tests** – the total number of SPT records currently displayed.
- **Maximum Depth** – the deepest SPT record in the current selection.

These values update automatically when filters are applied.

SPT Results Table

The table lists the SPT records used in the plot.

It may include:

- Location ID.
- Elevation.
- N-value.
- Geological legend.
- Geological unit.
- Other available AGS fields.

This table is useful when the user wants to inspect the actual values behind the plotted points.

Notes Section

The notes section provides important interpretation notes.

For example, the dashboard may state that values shown in the table are raw N-values unless corrected values are specifically selected.

Users should check this note before using the results for interpretation or reporting.

Atterberg Limits Page

The **Atterberg Limits** page displays Liquid Limit and Plasticity Index results from laboratory testing.

It allows users to view test results on a standard plasticity chart and compare results by geological unit.

Screenshot Placeholder – Atterberg Limits Page

Paste screenshot here:

[Atterberg Limits page](#)

Plasticity Chart

The main chart plots:

- **X-axis:** Liquid Limit (%)
- **Y-axis:** Plasticity Index (%)

Each point represents one Atterberg Limits test.

The points are coloured by interpreted geology, allowing users to see how plasticity varies between geological units.

Static Plasticity Chart Background

The plasticity chart includes a static background image showing standard classification boundaries.

This includes features such as:

- A-line.
- Clay and silt regions.
- Low, intermediate, high, very high and extremely high plasticity zones.

The background is provided as a visual aid. The dashboard plots the AGS data onto the chart but does not replace engineering interpretation.

Summary Cards

The summary cards provide an overview of the filtered Atterberg Limits data.

They typically include:

- **Average Liquid Limit**
- **Number of Tests**
- **Average Plasticity Index**

These update automatically as filters are applied.

Atterberg Limits Table

The table lists the laboratory records used in the plot.

It typically includes:

- Location ID.
- Liquid Limit.
- Plasticity Index.
- Sample top depth.
- Geological unit.

This allows users to check the source values behind the chart.

Units Panel

The units panel confirms the variables and units used on the plot, such as Liquid Limit (%) and Plasticity Index.

Geology Summary Page

The **Geology Summary** page provides a project-wide overview of the geological units recorded in the AGS dataset.

It is useful for checking stratigraphy, reviewing layer thicknesses and comparing top elevations across boreholes.

Screenshot Placeholder – Geology Summary Page

Paste screenshot here:

[Geology Summary page](#)

GEOL_GEOL Stratigraphy Summary

This table summarises the primary geology codes from the `GEOL_GEOL` field.

For each geology code, the table may show:

- Minimum layer thickness.
- Average layer thickness.
- Maximum layer thickness.
- Minimum top elevation.
- Average top elevation.
- Maximum top elevation.

This gives a quick overview of the thickness and elevation range of each primary geological unit.

GEOL_GE02 Stratigraphy Summary

This table provides a similar summary using the interpreted geology field, `GEOL_GE02`.

This is helpful where `GEOL_GE02` has been used to simplify or standardise geology descriptions across the dataset.

Stratigraphy by Hole – GEOL_GEOL Top Elevations

This table compares geology across boreholes using the primary geology code.

Each row represents a location ID, and each column represents a geology code.

The values show the top elevation at which that geological unit is encountered.

This helps users compare the relative level of strata across the site.

Stratigraphy by Hole – GEOL_GEO2 Top Elevations

This table provides the same type of comparison using the interpreted geology field.

It is useful for quickly reviewing simplified stratigraphic trends across the site.

Geological Map

The map shows the investigation locations, coloured by geology.

This allows users to review the spatial distribution of geological units across the project area.

Undrained Shear Strength Cu Page

The **Undrained Shear Strength Cu** page compares measured undrained shear strength values with estimated values derived from SPT correlations.

This allows users to review laboratory triaxial test results alongside empirical estimates from SPT data.

Screenshot Placeholder – Undrained Shear Strength Cu Page

Paste screenshot here:

Undrained Shear Strength Cu page

Measured Cu from Triaxial Tests

The left-hand plot shows measured undrained shear strength values from triaxial testing.

Each point represents a laboratory test result, typically extracted from the TRIT table.

The plot shows:

- **X-axis:** Measured Cu, usually in kPa.
- **Y-axis:** Sample depth or sample elevation.

The points are coloured by geology.

Estimated Cu from SPT Correlation

The right-hand plot shows estimated undrained shear strength values calculated from SPT results.

These values are based on an empirical correlation using the selected **F1 factor**.

This allows users to compare estimated Cu values with measured laboratory results where both datasets are available.

F1 Factor

The **F1 factor** control allows the user to adjust the correlation used to estimate Cu from SPT data.

Changing the F1 factor updates the estimated Cu plot automatically.

For example, where the dashboard uses a relationship such as:

Estimated Cu = F1 × SPT N-value

increasing the F1 factor will increase the estimated Cu values shown on the plot.

Important Estimated Cu values are indicative and depend on the selected empirical factor. They should be used with engineering judgement and compared with measured laboratory data where available.

Axis Selection

Both plots allow the Y-axis to be changed between depth and elevation.

This allows users to review strength values either relative to ground level or relative to a common elevation datum.

Rock & Material Properties Page

The **Rock & Material Properties** page displays laboratory rock and material test results where these are present in the AGS file.

This page may appear empty if the example AGS dataset or user-selected AGS file does not contain rock testing data.

Screenshot Placeholder – Rock & Material Properties Page

Paste screenshot here:

[Rock and Material Properties page](#)

Uniaxial Compressive Strength

The Uniaxial Compressive Strength plot displays UCS results, typically from the `RUCS` table.

The plot shows UCS values against either sample depth or sample elevation.

Point Load Strength Index

The Point Load plot displays point load test results, typically from the `RPLT` table.

It allows users to review point load strength index values against depth or elevation.

Dry Density

The Dry Density plot displays dry density values, typically from the relevant density testing table such as `RDEN`.

Results can be viewed against sample depth or sample elevation.

Moisture Content

The Moisture Content plot displays moisture content results from laboratory testing.

Results can also be viewed against depth or elevation.

Axis Selection

Each chart includes a selector allowing the user to choose whether the vertical axis shows:

- Sample Depth
- Sample Elevation

Dynamic Units

The units table beneath the plots shows the units and axis labels for the displayed test results.

These are read from the AGS data where available.

Particle Size Distribution Page

The **Particle Size Distribution** page displays laboratory particle size distribution curves.

It allows users to compare grading curves between samples, boreholes and geological units.

Screenshot Placeholder – Particle Size Distribution Page

Paste screenshot here:

[Particle Size Distribution page](#)

PSD Chart

The main chart plots PSD curves using data from AGS particle size distribution tables, such as GRAG and GRAT.

The chart shows:

- **X-axis:** Particle size, usually in millimetres.
- **Y-axis:** Percentage passing.

Each line represents an individual PSD test.

The legend identifies each curve using the location ID and sample depth, allowing users to identify which sample each curve relates to.

Static PSD Background

The chart uses a static Particle Size Distribution background image as an underlay.

This background provides familiar grading chart context, including:

- Clay.
- Silt.
- Sand.
- Gravel.
- Cobbles.
- Standard sieve size divisions.

The plotted lines are overlaid onto this background.

Note The PSD background is a visual guide. The dashboard plots the imported AGS laboratory data onto the chart but does not alter or reinterpret the source data.

Filtering PSD Curves

PSD charts can become difficult to read when many curves are shown at once.

For clearer review, users should filter by:

- Location ID.
- Geological unit.
- Location type.
- Source file.
- Project ID.

This allows smaller groups of curves to be compared more clearly.

7.5 Troubleshooting and Useful Checks

Before relying on the dashboard outputs, users should carry out a few simple checks.

- Confirm that the folder name, project ID, project name and source file are correct on the Project Summary page.
- Use the AGS Data Loaded table to confirm which AGS groups were imported and whether they contain records.
- Review the units panels on each page, especially before using plots for reporting or interpretation.
- If charts are difficult to read, apply filters to focus on specific boreholes, geology codes or location types.
- If a page has no data, check whether the relevant AGS table exists and contains records.

Troubleshooting

Empty pages

Some pages may appear empty if the imported AGS file does not contain the relevant test data.

This does not necessarily mean the dashboard has failed.

For example:

- If there are no SPT records, the **SPT Analysis** page may be empty.
- If there are no Atterberg Limits records, the **Atterberg Limits** page may be empty.
- If there are no rock testing records, the **Rock & Material Properties** page may be empty.
- If there are no PSD records, the **Particle Size Distribution** page may be empty.

The **Project Summary** page can be used to check which AGS groups have been imported and whether data is present.

The Dashboard Opens but Charts Are Empty

Check that:

- The correct folder path was entered.
- The folder contains the generated CSV files.
- The files have not been renamed.
- The AGS to CSV conversion completed successfully.
- The relevant AGS tables contain data.

The Wrong Project Appears

Check that the FolderPath points to the intended CSV output folder.

If needed, reopen the template or edit the parameter and load the correct folder.

Some Pages Are Empty

This usually means the AGS dataset does not contain that type of data.

For example, the Rock & Material Properties page will be empty if there are no UCS, point load, density or moisture content records.

A Plot Looks Too Busy

Apply filters at the bottom of the page to reduce the number of displayed records.

For example, filter by one borehole, one geological unit or one source file.

Units Look Unexpected

Check the units stored in the source AGS file and review the dynamic units table on the relevant page.

7.6 Output

Include note here about how to output charts etc to QGIS:Layouts/Word etc

8. Help and feedback

8.1 Links to General third-party documentation and information

Links to General guidance

- [QGIS documentation](#)
- [PowerBI documentation](#)
- [AGS Data Format](#)
- [GeoPackage](#)
- [GDMS-Geodatadets](#)
- [BGS mapping Layers](#)
- [BGS GeoIndex](#)

Links to useful QGIS guidance

- [Overview of the interface](#)
- [QGISProjects](#)
- [managing plugins](#)
- [Installing plugins from zip file](#)
- [Layout Manager](#)
- [Editing Layouts](#)
- [Layouts - map item](#)
- [Layouts - image item](#)
- [Attribute tables](#)
- [Attribute tables -edit data](#)
- [layer symbology](#)
- [layer labelling](#)
- [layer transparency](#)
- [Processing Toolbox](#)
- [Setting Relationships](#)
- [Selecting features](#)
- [Identifying Features](#)
- [Navigating the map](#)
- [Point clouds](#)
- [Elevation Profile](#)

Links QGIS plugin guidance

- [DataPlotly](#)
- [OpenLog](#)
- [Select By Relationship](#)

8.2 Feedback

Please use the following link to a form, to provide feedback on this documentation and the tools used (PowerBI and QGIS). Please be as detailed as possible when requesting wish-list items or bugs/problems.

[Insert link to form](#)

9. Guidance for IT

Including versions of software.

10. FAQs

Select an item from the list on the right.

10.1 Convert an AGS file to GeoPackage

In QGIS: select the ags-tools processing plugin AGS2DB...

10.2 Edit geology codes

In QGIS: select the GEOL layer - view attribute table - toggle editing, change/add data, save

10.3 Create an exploratory hole plan

In QGIS: select the layers required on the map. Select File-Layout Manager....

10.4 View summary plots such as SPT vs depth

In QGIS: select the ags-tools processing plugin - Summary

10.5 Plot and analyse data

In QGIS: select the ags-tools processing plugin - DB2CSV or AGS2CSV

10.6 Create graphical exploratory hole logs



Tip

GeoPackage file must be loaded in QGIS

In QGIS: select OpenLog. Follow the guidance here ([link to OpenLog plugin guidance](#))

10.7 Filter data by selecting exploratory holes on a map



Tip

Relationships in QGIS are required.

In QGIS: ensure Relationships are set and Select By Relationship plugin is installed and selected. See guidance here ([link to Select By Relationship plugin guidance](#))

10.8 Create a cross section

 Tip

Section lines can be set by co-ordinates, or drawn by hand

In QGIS: select ags-tools processsing plugin 'cross-sections'. See guidance here ()